Smart Plastic Bottle Recycling Machine with AI and Mobile Money Payments

Project Overview

The project is to develop a smart plastic bottle recycling machine that allows people to deposit plastic bottles and receive payments directly through mobile money. The machine will use a keypad, sensors, and a camera connected to a Raspberry Pi running AI software to identify genuine plastic bottles and reject other objects, such as wood, to prevent fraud.

How It Works

1. Enter the quantity: The user enters the expected number of bottles using the machine's keypad.
2. Insert bottles: The user inserts bottles one at a time through the machine's entry point.
3. Detect and count: A sensor detects each inserted object and helps prevent duplicate counting.
4. AI bottle verification: The Raspberry Pi uses the camera and locally running AI software to analyze each object and determine whether it is an acceptable plastic bottle. Unaccepted objects are rejected and do not qualify for payment.
5. Calculate the reward: The ESP32-WROOM records the number of accepted bottles and calculates the payment. If the user enters 10 bottles but only 8 are accepted, the machine pays for 8 bottles.
6. Receive mobile money: The user enters their mobile money phone number and confirms the transaction. The system sends the calculated payment through the supported mobile money service.
7. Receive an SMS confirmation: The user receives a text message confirming the transaction, including the number of accepted bottles, any rejected bottles, the amount paid, and the transaction reference, where supported.
8. Collect bottles: Accepted bottles are stored in the machine for collection and recycling.

Main Components

- Raspberry Pi and camera: Run AI software offline to identify acceptable plastic bottles and reject other objects.
- ESP32-WROOM: Controls the machine, reads the keypad and sensors, and manages the accepted bottle count.
- Keypad: Allows users to enter the expected bottle quantity and mobile money number.
- Infrared or photoelectric sensor: Detects each item entering the machine.
- Display: Shows the number of bottles accepted, rejected, and the payment amount.
- Mobile money integration: Processes payments to users' phone numbers.

- SMS notification service: Sends transaction confirmations showing the accepted and rejected bottle counts.
- Bottle entry mechanism and collection compartment: Help control item entry and store accepted bottles securely.

Key Features

- AI-powered bottle verification to reduce fraud.
- Offline image recognition, reducing dependence on internet data for AI processing.
- Automatic counting and verification of each item.
- Mobile money payments directly to users.
- SMS receipts confirming accepted and rejected bottle quantities and payment details.
- Transparent reward calculations based on verified bottles.